

Activating and Retaining Marketplace Sellers

Executive Summary — Data/Product Analyst Portfolio

BOTTOM LINE

Just 14.9% of sellers (438, "Top Sellers") generate 69.6% of marketplace GMV. Right-censoring-corrected: sellers making 6+ sales in their first 30 days stay active the following month at 94.6% (vs. 34.5% for one-sale sellers). GMV falls 41.2% on average in the 3 months before a seller churns. A bootstrap 95% CI confirms buyer-seller activity move together within the same month ($r=0.987$, CI [0.94, 1.00]) — but whether one predicts the other a month ahead is genuinely inconclusive ($r=0.031$, 95% CI [-0.79, 0.57], $n=18$ monthly pairs), not "no relationship" as first reported.

SELLERS ANALYZED

2,945

20-month window

TOP-SEGMENT GMV

69.6%

From just 14.9% sellers

HABIT → RETENTION

35% → 95%

1 sale vs. 6+ sales (corrected)

PRE-CHURN GMV DROP

-41.2%

Over last 3 active months

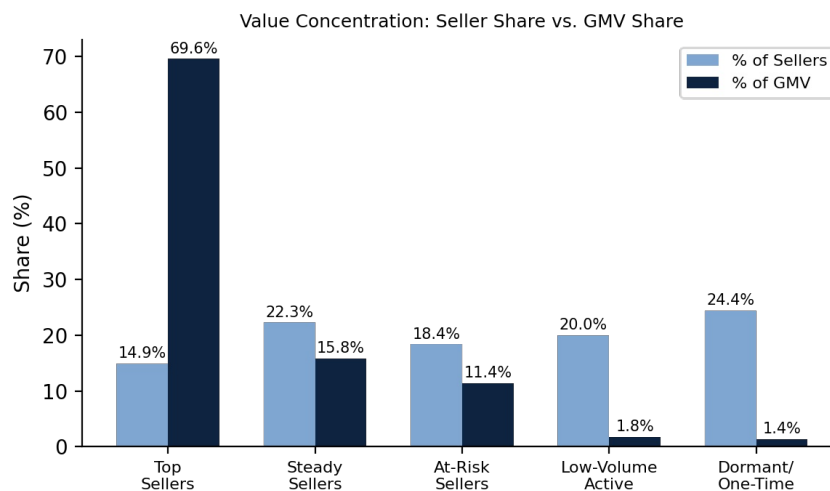
The Business Question

A marketplace has two kinds of "customers": buyers and sellers. Most retention analyses cover only the buyer side — this project addresses the less-common angle: the dynamics of sellers staying on or leaving the platform. The question: which sellers are most valuable, which are at risk of churning, and how can churn be detected early enough to act on it?

Methodology: Retention Lifecycle Framework

Built on the Amplitude Product Analytics Playbook: Critical Event (First Delivered Order) and Usage Interval (data-derived 80th-percentile gap of 42 days, mapped to the Playbook's Monthly bucket / 30-day threshold) were identified first. Each seller-month was then classified as New / Current / Resurrected / At Risk / Churned. RFM + K-means ($k=5$) produced five seller segments. Every finding below was stress-tested for methodological soundness — this revision documents issues a critical review caught and corrected with real recomputation (see Page 2 and 3).

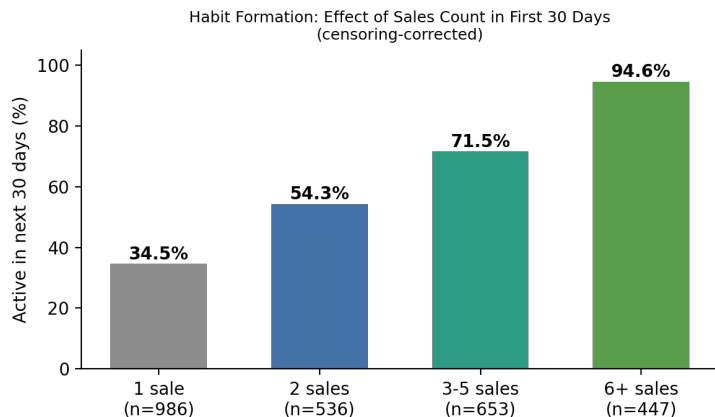
5 Seller Segments: GMV Concentration



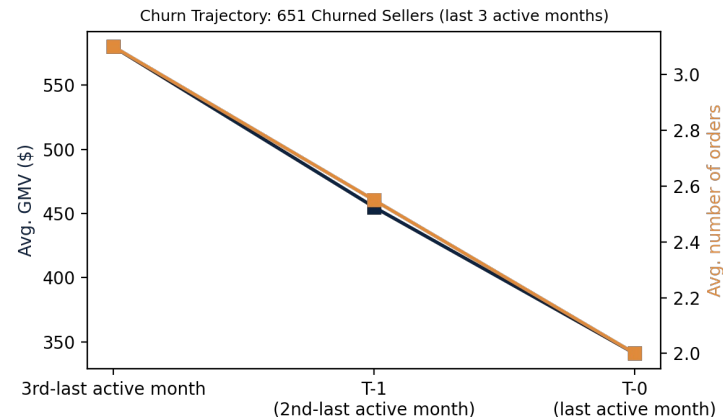
What We Found — and What We Got Wrong the First Time

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Habit Formation: An Early Signal



Churn Trajectory: A Gradual Decline



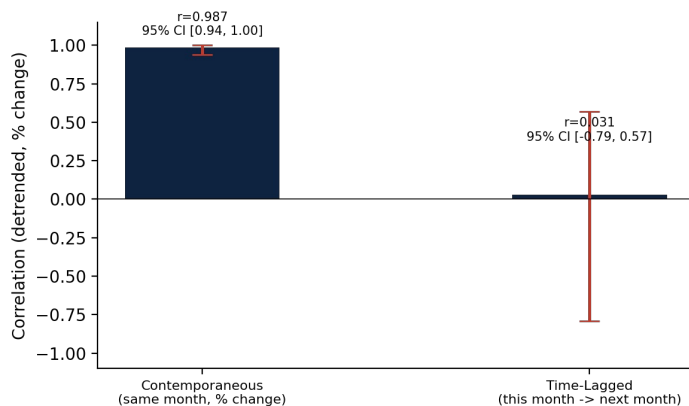
A near-linear relationship: the more a seller sells in their first 30 days, the more likely they stay active the next month. Correlation, not causation — this is not proof of a causal mechanism.

651 churned sellers: GMV -41.2%, orders -35.6% over their last 3 active months. Churn is not sudden — it is a gradual decline. Note: 535 sellers (18%) never made a second sale at all and are not in this chart — a distinct, more common churn pattern (see Dashboard, page 2).

CORRECTED — Right-Censoring in Habit Formation

323 of 2,945 sellers (11%) were excluded for lacking a full 60-day follow-up window before the dataset's 2018-08-31 end date. Corrected retention rates rose in every bucket: 1 sale 30.4%→34.5%, 2 sales 48.8%→54.3%, 3-5 sales 66.6%→71.5%, 6+ sales 89.1%→94.6%. The pattern held up and got slightly stronger — the original figures had understated retention, not overstated it.

Methodological Catch: Confidence Intervals Change the Conclusion



96.58% of buyers purchase from only one seller — so this analysis was run at monthly marketplace-aggregation level, not individual pairs. After adding a 1,000-resample bootstrap: the same-month correlation ($r=0.987$) has a tight 95% CI of [0.944, 0.996] — confirmed statistically real. The one-month-lagged correlation ($r=0.031$) has a 95% CI of [-0.792, 0.568] — wide and crossing zero. Revised conclusion: buyer and seller activity clearly move together within a month. Whether one predicts the other a month ahead cannot be determined from only 18 months of data — this is inconclusive, not "no relationship," which is what the uncorrected version claimed.

How to Use This — and What a Critical Review Fixed

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CORRECTED — All Flagged Issues Were Re-Run on Real Data

(1) Habit Formation right-censoring: after excluding sellers without a full 60-day follow-up window, retention rates rose in every bucket (34.5% to 94.6% across the four groups) — originals had understated retention. (2) Lifecycle Framework right-censoring: excluding the last 2 months (needed to confirm "Churned" status), the distribution shifted from Churned 35.97%/Current 35.31% to Current 36.44%/Churned 33.91% — Current, not Churned, is now the largest category. (3) Confidence intervals: a 1,000-resample bootstrap confirms the same-month buyer-seller correlation is real ($r=0.987$, 95% CI [0.944, 0.996]), but shows the one-month-lagged correlation is genuinely inconclusive ($r=0.031$, 95% CI [-0.792, 0.568], $n=18$ monthly pairs) rather than proof of "no relationship." (4) Engagement Trigger proxy: the "6.5x higher GMV" figure for Fast Responders was computed on average GMV, which is skewed by outliers; on median GMV — the more reliable comparison — the gap is 4.4x (\$1,508 vs. \$342). The "87%" figure is 87% of measurable sellers (those with 2+ sales), not 87% of all sellers (73.4% of the full 2,945).

Practical Applications

- New Seller Activation** — Push new sellers to reach 3-6 sales in their first 30 days — per the corrected Habit Formation finding, this could raise next-period retention from ~35% to ~95%.
- At-Risk Intervention** — Monitor sellers whose GMV/orders decline 35-40% in real time and intervene early — the trigger point from the Churn Trajectory finding.
- One-Time Seller Onboarding** — Since 18% of sellers never make a second sale, invest in first-to-second-sale conversion specifically — a distinct problem from gradual-decline churn.

Where to Be Careful — Limitations

- The pilot period (2016-09 — 2016-12) was excluded (domain-informed decision) — the analysis window is limited to 20 months.
- Two-Sided analysis is marketplace-level only — individual seller-buyer pairing is weak (96.58% of buyers use a single seller).
- Habit Formation and Churn Trajectory are correlational findings, not a predictive model. The Power User Curve / Value Concentration overlap (page 4) is an association, not a prediction, and the two methods share the same underlying dataset rather than being independent.
- All right-censoring / confidence-interval / denominator corrections are now applied consistently across Habit Formation, Lifecycle Framework, Two-Sided Marketplace, and Engagement Trigger.
- All findings are specific to the Olist (Brazil) dataset — they should be validated against local data before being transferred to Uzun Market.

Recommended Next Step

New Seller Activation and At-Risk Intervention remain the top priorities, now backed by fully right-censoring-corrected, confidence-interval-aware, and denominator-checked numbers across all analyses. No open methodological items remain from the critical review — this version is ready for stakeholder review.

Part of a Rigor-Driven Analytics Portfolio

This project applies the Product Analytics Playbook (Amplitude) framework to a marketplace's seller side, a companion to the UCI Online Retail (customer retention, RFM, K-means) project. Each finding is presented alongside a methodological check — including, in this revision, an honest account of what a critical review found wrong.

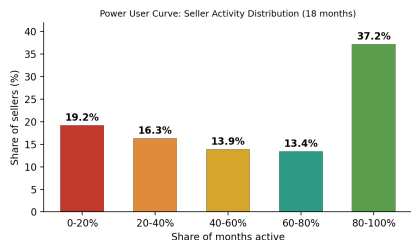
Engagement & Stickiness — A New Analytical Layer

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WHY THIS PAGE EXISTS

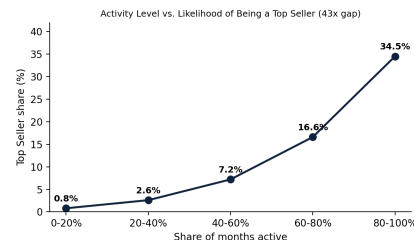
Applying the Amplitude 'Mastering Engagement' framework (Engagement Loop, Stickiness, Power User Curve) to this dataset surfaced two complementary confirmations of earlier findings and one genuinely new, useful negative result — without requiring any data we didn't already have. These are the same seller/order dataset viewed through a different analytical lens, not independent data sources.

Power User Curve: Stickiness Confirms Value



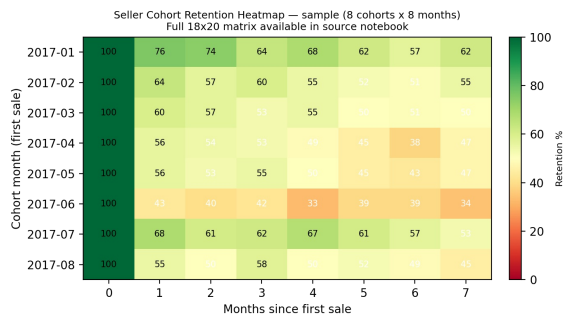
A U-shaped curve: 37.2% of sellers are active in 80-100% of months (a stable core), 19.2% in only 0-20% — a polarized base, the hallmark of a 'sticky' product per the framework.

...and It Is Strongly Associated With Top-Seller Status



Sellers active 80-100% of months are Top Sellers at 34.5% vs. 0.8% for the least active — a 43x gap. This is a correlation, not a predictive model: both come from the same GMV-based Top Seller definition, so it independently confirms Value Concentration through a different lens rather than predicting an outcome.

Cohort Retention Heatmap (sample)



Two Findings From the Engagement Lens

Geographic retention (Cohort x State): a modest 49.5%-57.0% range across states — a useful negative result showing churn is driven by seller-level behavior, not geography, so region-targeted spend is not the priority. Only states with 30+ qualifying sellers are shown; among all 2,053 qualifying sellers, SP alone accounts for 59% (1,218).

Engagement trigger proxy (inter-sale gap, 30-day threshold — mapped from the data-derived 42-day 80th percentile): Fast Responders (≤ 30 days between sales, 87% of measurable sellers) have 4.4x higher median GMV (\$1,508 vs. \$342; 6.5x on average GMV, which is skewed by outliers) and about half the churn rate of Slow Responders — consistent with, and a second confirmation of, the Habit Formation finding via the Engagement Loop's 'internal trigger' concept, rather than a fully independent discovery.

The 2017-06 cohort (heatmap, left) retains noticeably worse than neighboring months — worth a root-cause check before generalizing other findings to it.

Honest Framing

This page uses the Amplitude 'Mastering Engagement' vocabulary (trigger, stickiness, power user) as an analytical lens on existing order data, not as literal event tracking — the dataset has no notification or session logs. Where a metric substantially overlaps an earlier finding (the trigger proxy vs. Habit Formation), or where two analyses share the same underlying dataset, that overlap is stated plainly rather than presented as a new or independent discovery.